TWO INSTABILITY INTERVALS FOR A SWITCHED PLANAR SYSTEM WITH ONE NON-HURWITZ MATRIX

YITZCHAK SHMALO

ABSTRACT. Let $\dot{X}_t = A_{I_t} X_t$ in $\mathbb{R}^2$, where I jumps between 0 and 1 at rate r. Malrieu asked whether A_0, A_1 can be chosen so that the set of unstable rates is a union of several intervals. For the Hurwitz matrix $A_0 = \left(\begin{smallmatrix} -3/4 & -1 \\ 1 & -3/4 \end{smallmatrix}\right)$ and the non-Hurwitz matrix $A_1 = \operatorname{diag}(5/4, -11/4)$ we prove that the top Lyapunov exponent $\lambda(r)$ is continuous, positive on $(0, 1/10]$, negative on $[4.997, 5.003]$ and positive on $[19.95, 20.05]$, so the set of unstable rates has at least two components. The signs near $r = 5$ and $r = 20$ come from trigonometric polynomials with exact rational coefficients that solve the Poisson equation for the direction $X_t/|X_t|$ up to an explicit error. Numerically λ changes sign twice, near $r = 1.2228$ and $r = 7.7616$. The case in which both matrices are Hurwitz, as in the example behind the question, remains open.

1. INTRODUCTION

Let A_0, A_1 be real 2×2 matrices and $r > 0$. Let $(I_t)_{t \geq 0}$ be the Markov process on $\{0, 1\}$ that jumps from 0 to 1 and from 1 to 0 at rate r, and let X solve $\dot{X}_t = A_{I_t} X_t$. Equivalently, (X, I) is the Markov process on $\mathbb{R}^2 \times \{0, 1\}$ with generator

$$Lf(x, i) = (A_i x) \cdot \nabla_x f(x, i) + r\big(f(x, 1 - i) - f(x, i)\big). \tag{1}$$

Let Φ_t solve $\dot{\Phi}_t = A_{I_t} \Phi_t$, $\Phi_0 = \mathrm{Id}$. The top Lyapunov exponent is $\lambda(r) = \lim_{t \to \infty} t^{-1} \log \|\Phi_t\|$ when this limit exists almost surely and is not random; for the pair of Theorem 1.1 it does (Proposition 2.1). Call a rate r *unstable* if $\lambda(r) > 0$ and *stable* if $\lambda(r) < 0$.

This is the setting of Section 4.1 of Malrieu's survey [9], which presents an example of Lawley, Mattingly and Reed [8]. For $\alpha > 0$ the matrices $A_0 = \left(\begin{smallmatrix} -\alpha & 1 \\ 0 & -\alpha \end{smallmatrix}\right)$ and $A_1 = \left(\begin{smallmatrix} -\alpha & 0 \\ -1 & -\alpha \end{smallmatrix}\right)$ are Hurwitz (all eigenvalues have negative real part), and so is every convex combination of them. Yet $|X_t| \to \infty$ almost surely for some α and some rates r. For this family the set of unstable rates is empty or, under Conjecture 4.3 of [9], an interval. Malrieu then asks the following.

Open question 3 of [9] (Oscillations of the Lyapunov exponent). Is it possible to choose the two 2×2 matrices A_0 and A_1 in such a way that the set of jump rates r associated to unstable processes is the union of several intervals?

The question is stated for two real 2×2 matrices. Theorem 1.1 answers it with a pair in which A_0 is Hurwitz and A_1 is not; for two Hurwitz matrices it is open (Section 6).

Theorem 1.1. *Let*

$$A_0 = \begin{pmatrix} -3/4 & -1 \\ 1 & -3/4 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 5/4 & 0 \\ 0 & -11/4 \end{pmatrix}. \tag{2}$$

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Then λ is continuous on $(0,\infty)$, and for every $X_0 \neq 0$ and every $r > 0$, $t^{-1}\log|X_t| \to \lambda(r)$ almost surely. Moreover

$$\begin{aligned} \lambda(r) &> \tfrac{9}{50} && \text{for } 0 < r \le \tfrac{1}{10}, \\ -0.0756 \le \lambda(r) &\le -0.0055 && \text{for } 4.997 \le r \le 5.003, \\ 0.0361 \le \lambda(r) &\le 0.1192 && \text{for } 19.95 \le r \le 20.05. \end{aligned}$$

Consequently the set $U = \{r > 0 : \lambda(r) > 0\}$ of unstable rates is open and has at least two connected components, one containing $(0, 1/10]$ and another containing $[19.95, 20.05]$, and they are separated by the stable interval $[4.997, 5.003]$.

The matrix A_0 has eigenvalues $-\frac{3}{4} \pm i$ and is Hurwitz; A_1 has eigenvalues $\frac{5}{4}$ and $-\frac{11}{4}$ and is not. Two limits of λ are positive. As $r \to 0$, $\lambda(r) \to \frac{1}{2}\big(-\frac{3}{4} + \frac{5}{4}\big) = \frac{1}{4}$ (Proposition 4.1), and as $r \to \infty$, $\lambda(r)$ tends to the largest eigenvalue $-\frac{3}{4} + \frac{\sqrt{3}}{2}$ of $\bar{A} = (A_0 + A_1)/2$ [4, Proposition 6.4]. The set U therefore has at least two components as soon as $\lambda(r) < 0$ at one rate. Neither limit gives the sign of λ at a rate of order one, and the direct sums of [8], which give k components in dimension $2k$, need dimension at least four.

The sign at $r = 5$ comes from the following estimate. Write $X_t = |X_t|(\cos\Theta_t, \sin\Theta_t)$, so that $\frac{d}{dt}\log|X_t| = h_{I_t}(\Theta_t)$ and $\dot{\Theta}_t = v_{I_t}(\Theta_t)$ with $h_0 = -\frac{3}{4}$, $h_1(\theta) = -\frac{3}{4} + 2\cos 2\theta$, $v_0 = 1$ and $v_1(\theta) = -2\sin 2\theta$; the pair (Θ, I) is the *angular process*. For every $r > 0$ and every pair $f = (f_0, f_1)$ of C^1 functions of period π,

$$\min_{i,\theta} g_i(\theta) \;\le\; \lambda(r) \;\le\; \max_{i,\theta} g_i(\theta), \qquad g_i = h_i + v_i f_i' + r(f_{1-i} - f_i)$$

(Lemma 3.1). A pair f with $g_0 = g_1 = \lambda(r)$ would solve the Poisson equation of the angular process; a pair of trigonometric polynomials with rational coefficients is called a *certificate* at rate r. For such a pair g_0 and g_1 are trigonometric polynomials with rational coefficients, and g_i differs from its constant coefficient by at most the sum of the absolute values of the others. With f of degree $2K$ this gives $-0.0714 \le \lambda(5) \le -0.0097$ for $K = 128$ and $0.0703 \le \lambda(20) \le 0.0850$ for $K = 32$ (Proposition 3.2); the second bound gives an explicit unstable rate above $r = 5$, while the limit $r \to \infty$ gives instability only beyond a rate it does not quantify. Since g_i depends on r only through the term $r(f_{1-i} - f_i)$, the same two pairs give the signs on $[4.997, 5.003]$ and $[19.95, 20.05]$ (Corollary 3.3). By the almost sure limit in Theorem 1.1, Malrieu's set $\{r : |X_t| \to \infty \text{ almost surely}\}$ contains U and contains no stable rate, so it is not an interval either.

Related work. Lawley, Mattingly and Reed [8] prove that switching between Hurwitz matrices is stable at slow and fast rates when the averaged matrix is Hurwitz. In their Example 3.3 they take direct sums of planar blocks to build pairs of $2k \times 2k$ matrices whose set of unstable rates has k separated components. The smallest case with two components is in dimension four. Lagasquie [7] studies a planar pair of the same kind, two Hurwitz matrices all of whose convex combinations are Hurwitz, and observes numerically a second region of instability after replacing the exponential holding times by Gamma holding times; his model keeps two planar matrices but changes the switching law. Benaïm, Le Borgne, Malrieu and Zitt [1] take two planar Hurwitz matrices with a convex combination that has a positive eigenvalue and show that $|X_t|$ tends to 0 or to infinity according to the rate. Porfiri, Jeter and Belykh find disconnected windows of stability for jump linear systems in discrete time, where an independent matrix is held for an integer number of steps [14], and antiresonance in a nonlinear planar system that switches between unstable modes [15]. Neither setting is the two-state exponential switching

of (1). Benaïm and Strickler [3] relate the top exponent of switched linear systems to invariant measures of the angular process and treat fast switching; Chitour, Mazanti, Monmarché and Sigalotti [4] compare deterministic and probabilistic exponents and give the fast-switching limit used in Sections 1 and 5. Slow and fast limits for cooperative systems are in [2] and [11], and a three-dimensional cooperative example with a stable convex hull and unstable switching is in [12]. None of these gives two planar matrices whose set of unstable rates has two components under the switching law (1).

Section 2 sets up the angular process and proves that λ exists and is continuous. Section 3 proves the certificate lemma and the signs near $r = 5$ and $r = 20$, and Section 4 the sign at small rates together with the proof of Theorem 1.1. Section 5 computes the whole curve from the stationary density, and Section 6 proves that a pair of normal Hurwitz matrices has no unstable rate.

2. The angular process and the exponent

For $\theta \in \mathbb{R}$ put $u_\theta = (\cos\theta, \sin\theta)$ and $u_\theta^\perp = (-\sin\theta, \cos\theta)$. For a matrix A let $h_A(\theta) = u_\theta^{\mathsf{T}} A u_\theta$ and $v_A(\theta) = (u_\theta^\perp)^{\mathsf{T}} A u_\theta$. Both h_A and v_A have period π, and Θ_t below is taken modulo π. Writing $X_t = R_t u_{\Theta_t}$ with $R_t = |X_t|$, the equation $\dot X = A_I X$ becomes

$$\frac{d}{dt}\log R_t = h_{I_t}(\Theta_t), \qquad \dot\Theta_t = v_{I_t}(\Theta_t), \tag{3}$$

with $h_i = h_{A_i}$ and $v_i = v_{A_i}$. The angular process (Θ, I) is a Markov process on $(\mathbb{R}/\pi\mathbb{Z}) \times \{0,1\}$ whose generator acts on pairs $f = (f_0, f_1)$ of C^1 functions with period π by

$$(\mathcal{L}_r f)_i = v_i f_i' + r(f_{1-i} - f_i). \tag{4}$$

For the matrices (2),

$$v_0 = 1, \qquad v_1(\theta) = -2\sin 2\theta, \qquad h_0 = -\tfrac{3}{4}, \qquad h_1(\theta) = -\tfrac{3}{4} + 2\cos 2\theta. \tag{5}$$

In state 0 the direction rotates at unit speed and the norm decays at rate $3/4$; in state 1 the direction moves toward the horizontal axis.

Proposition 2.1. *For the matrices* (2) *and every* $r > 0$ *there is* $\lambda(r) \in \mathbb{R}$ *such that* $t^{-1}\log|X_t| \to \lambda(r)$ *almost surely for every* $X_0 \neq 0$ *and every initial state* I_0*;* $t^{-1}\log\|\Phi_t\| \to \lambda(r)$ *almost surely, and* λ *is continuous on* $(0,\infty)$.

Proof. Start the clock at a jump into state 1 and call a *cycle* a holding period T_1 in state 1 followed by the next holding period T_0 in state 0; the holding times are independent and exponential with mean $1/r$. Let $\Theta^{(n)}$ be the direction at the start of the n-th cycle; its two holding times $(T_1^{(n)}, T_0^{(n)})$ are independent of $\Theta^{(n)}$. During a period in state 0 the direction moves by T_0 modulo π, and T_0 modulo π has density

$$k_r(s) = \frac{re^{-rs}}{1 - e^{-\pi r}}, \qquad 0 \le s < \pi, \tag{6}$$

so $m_r \le k_r \le \overline{m}_r$ with $m_r = r/(e^{\pi r} - 1)$ and $\overline{m}_r = r/(1 - e^{-\pi r})$. Since $\Theta^{(n+1)}$ is the direction at the end of the period in state 1 shifted by $T_0^{(n)}$ modulo π, and $T_0^{(n)}$ is independent of that direction, the transition kernel of the chain $(\Theta^{(n)})$ is bounded below by $m_r\, d\theta$, uniformly in the starting direction. By Doeblin's condition the chain has a unique stationary law and converges to it in total variation at a geometric rate uniform in the starting point [10, Theorem 16.0.2].

The increment of $\log R$ over the n-th cycle is a function of $\Theta^{(n)}$ and of $(T_1^{(n)}, T_0^{(n)})$, bounded in absolute value by $\kappa(T_0^{(n)} + T_1^{(n)})$, with $\kappa = \max_i \|A_i\|$; this bound is integrable. The

chain $(\Theta^{(n)}, T_1^{(n)}, T_0^{(n)})$ has two-step transition kernel bounded below by $m_r\, d\theta \otimes \mathrm{Law}(T_1, T_0)$, because $\Theta^{(n+2)}$ is a function of $(\Theta^{(n+1)}, T_1^{(n+1)})$ shifted by the independent $T_0^{(n+1)}$ modulo π, and is independent of $(T_1^{(n+2)}, T_0^{(n+2)})$. It is therefore uniformly ergodic, and the law of large numbers [10, Theorem 17.0.1] applies to the cycle increment. Together with the strong law of large numbers for the independent sums $\sum_n (T_0^{(n)} + T_1^{(n)})$ it gives $t^{-1} \log R_t \to \lambda(r)$ along cycle ends, with $\lambda(r)$ equal to the stationary mean of the cycle increment divided by the mean cycle length $2/r$. Between cycle ends $\log R$ moves by at most $\kappa(T_0 + T_1)$ for the current cycle, and $\max_{n \le N}(T_0^{(n)} + T_1^{(n)})/N \to 0$ almost surely, so the limit holds along all t. It does not depend on $X_0 \neq 0$; applying it to two linearly independent initial vectors gives $t^{-1} \log \|\Phi_t\| \to \lambda(r)$.

For continuity, fix a compact interval $[a, b] \subset (0, \infty)$. The constant m_r is bounded below on $[a, b]$, so the Doeblin bound is uniform there. The holding times have joint density $r^2 e^{-r(t_0+t_1)}$, continuous in r in L^1, while the deterministic maps from the holding times to the next direction and to the cycle increment do not depend on r. Hence the transition kernel is continuous in r in total variation, uniformly in the starting direction, and by the uniform geometric contraction so is the stationary law. The conditional mean of the cycle increment given the starting direction is bounded and continuous in r uniformly in the direction, by dominated convergence with the bound $\kappa(t_0 + t_1)$. Dividing the stationary mean by $2/r$ gives continuity of λ on $[a, b]$. □

3. Certificates

Lemma 3.1. *For the matrices* (2)*, let $r > 0$ and let $f = (f_0, f_1)$ be a pair of C^1 functions of period π. Put $g_i = h_i + (\mathcal{L}_r f)_i$. If $\ell \le g_i(\theta) \le u$ for $i = 0, 1$ and all θ, then $\ell \le \lambda(r) \le u$.*

Proof. By Dynkin's formula, $N_t = f_{I_t}(\Theta_t) - f_{I_0}(\Theta_0) - \int_0^t (\mathcal{L}_r f)_{I_s}(\Theta_s)\, ds$ is a martingale. Its jumps occur at the jumps of I, at rate r, and are bounded by $2\|f\|_\infty$, so $\mathbb{E} N_t^2 \le 4r\|f\|_\infty^2 t$ and $N_n/n \to 0$ almost surely along integers by the strong law for square-integrable martingales [5, Theorem 2.18]. Since $|N_t - N_n| \le 2\|f\|_\infty + \|\mathcal{L}_r f\|_\infty$ for $n \le t \le n+1$, also $N_t/t \to 0$. By (3),

$$\frac{1}{t} \log \frac{R_t}{R_0} = \frac{1}{t} \int_0^t g_{I_s}(\Theta_s)\, ds - \frac{f_{I_t}(\Theta_t) - f_{I_0}(\Theta_0) - N_t}{t},$$

and the last term tends to 0. The integral average lies in $[\ell, u]$, and the left side tends to $\lambda(r)$ by Proposition 2.1. □

Take $f_i(\theta) = c_{i,0} + \sum_{k=1}^{K} \big(c_{i,k} \cos 2k\theta + s_{i,k} \sin 2k\theta\big)$ with rational coefficients. Since $v_0 = 1$ and $v_1 = -2 \sin 2\theta$, each g_i is a trigonometric polynomial of degree at most $2(K+1)$ with rational coefficients,

$$g_i(\theta) = C_i + \sum_{k=1}^{K+1} \big(\alpha_{i,k} \cos 2k\theta + \beta_{i,k} \sin 2k\theta\big), \tag{7}$$

and therefore

$$C_i - \sum_{k=1}^{K+1} \big(|\alpha_{i,k}| + |\beta_{i,k}|\big) \le g_i \le C_i + \sum_{k=1}^{K+1} \big(|\alpha_{i,k}| + |\beta_{i,k}|\big). \tag{8}$$

Proposition 3.2. *There are rational pairs $f^{(5)}$ with $K = 128$ and $f^{(20)}$ with $K = 32$, all coefficients with denominators dividing 10^{12}, for which the bounds* (8) *at $r = 5$ and $r = 20$ are the numbers in Table 1. Consequently*

$$-0.071355633951 \le \lambda(5) \le -0.009745897257, \qquad 0.070309824390 \le \lambda(20) \le 0.084999667350.$$

r	state i	lower bound in (8)	upper bound in (8)
5	0	−0.040550774017	−0.040550757183
5	1	−0.071355633951	−0.009745897257
20	0	0.077654745278	0.077654746482
20	1	0.070309824390	0.084999667350

TABLE 1. The bounds (8) for $g_i = h_i + (\mathcal{L}_r f)_i$, with $f = f^{(5)}$ at $r = 5$ and $f = f^{(20)}$ at $r = 20$. Every entry is a rational number with denominator dividing 10^{12}, written in full.

Proof of Proposition 3.2. The coefficients are listed in the repository, in the file `data/poisson_certificates.json`, whose SHA-256 checksum begins `737e1521068d8de6` (the full value is in `SHA256SUMS`). They are twelve-place decimal roundings of a floating-point solution of the Poisson equation restricted to the harmonics $k \le K$. Given the coefficients, (7) is computed by exact rational arithmetic, using $\frac{d}{d\theta}\cos 2k\theta = -2k\sin 2k\theta$, $\frac{d}{d\theta}\sin 2k\theta = 2k\cos 2k\theta$ and the product formulas for $\sin 2\theta\cos 2k\theta$ and $\sin 2\theta\sin 2k\theta$. The file `code/verify_certificate.py` carries out this computation with exact rationals in the basis $\cos 2k\theta, \sin 2k\theta$, and `code/certificate_laurent.py` in the basis z^k, $z = e^{2i\theta}$, where $v_1 = i(z - z^{-1})$; both give the entries of Table 1. Lemma 3.1 with $\ell = \min_i$ and $u = \max_i$ of the table entries gives the bounds. □

Corollary 3.3. $-0.0756 \le \lambda(r) \le -0.0055$ *for* $4.997 \le r \le 5.003$ *and* $0.0361 \le \lambda(r) \le 0.1192$ *for* $19.95 \le r \le 20.05$.

Proof. By (4), $(\mathcal{L}_r f)_i - (\mathcal{L}_{r_0} f)_i = (r - r_0)(f_{1-i} - f_i)$, so with the same f, at rate r each g_i lies between the bounds (8) at r_0 widened by $|r - r_0|\,\|f_1 - f_0\|_\infty$ on each side. Bounding $\|f_1 - f_0\|_\infty$ by the sum of the absolute values of the Fourier coefficients of $f_1 - f_0$ gives at most 1.3918 for $f^{(5)}$ and at most 0.6824 for $f^{(20)}$ (the exact rationals are in `data/certified_neighborhoods.json`). For $f^{(5)}$ the widened bounds for g_0 and g_1 lie in

$$[-0.071355633951 - 0.003 \cdot 1.3918,\ -0.009745897257 + 0.003 \cdot 1.3918] \subset [-0.0756,\ -0.0055],$$

and for $f^{(20)}$ they lie in

$$[0.070309824390 - 0.05 \cdot 0.6824,\ 0.084999667350 + 0.05 \cdot 0.6824] \subset [0.0361,\ 0.1192],$$

since by Table 1 the bounds for g_0 lie inside those for g_1 at both rates. Lemma 3.1 gives the bounds. □

4. Small rates

Proposition 4.1. *For every* $r > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{\pi \log 2}{2}\,\frac{r^2}{1 - e^{-\pi r}}. \tag{9}$$

In particular $\lambda(r) > 9/50$ *for* $0 < r \le 1/10$, *and* $\lambda(r) \to 1/4$ *as* $r \to 0$.

Proof. Use the cycles of the proof of Proposition 2.1. In state 0, $|e^{A_0 t}x| = e^{-3t/4}|x|$. In state 1, for a unit vector u_θ,

$$\log|e^{A_1 T_1}u_\theta| \ge \log\bigl(e^{5T_1/4}|\cos\theta|\bigr) = \tfrac{5}{4}T_1 + \log|\cos\theta|.$$

The direction $\Theta^{(n)}$ at the start of a cycle, where the period in state 1 begins, is the direction at the end of the previous period in state 1 shifted by an independent T_0 modulo π, so its

conditional density given the past is at most $\overline{m}_r$. Since $\log|\cos\theta| \le 0$ and $\int_0^\pi \log|\cos\theta|\,d\theta = -\pi\log 2$,

$$\mathbb{E}\big[\log|\cos\Theta^{(n)}|\,\big|\,\text{past}\big] \ge -\overline{m}_r\pi\log 2.$$

The stationary mean of the cycle increment is therefore at least $-\frac{3}{4r} + \frac{5}{4r} - \overline{m}_r\pi\log 2$, and dividing by the mean cycle length $2/r$ gives (9). Using $1 - e^{-x} \ge x/(1+x)$ for $x > 0$,

$$\lambda(r) \ge \frac{1}{4} - \frac{r(1+\pi r)\log 2}{2},$$

and for $0 < r \le 1/10$, with $\pi < 4$ and $\log 2 < 1$, the right side exceeds $\frac{1}{4} - \frac{1}{20}\cdot\frac{14}{10} = \frac{9}{50}$. For the limit, the cycle increment is at most $-\frac{3}{4}T_0 + \frac{5}{4}T_1$, so $\lambda(r) \le \frac{1}{4}$ for every r. □

Proof of Theorem 1.1. Continuity and the almost sure limit are Proposition 2.1. The bounds are Proposition 4.1 and Corollary 3.3. Since λ is continuous, U is open in $(0,\infty)$ and each of its connected components is an open interval. The points $1/10$ and 20 lie in U, and the point 5 between them does not, so they lie in different components. □

5. Numerical picture

The certificates prove signs at chosen rates; the shape of the whole curve is computed numerically. The exponent of the pair of [8] is computed from the stationary densities of the angular process, which are explicit [9, Theorem 4.1]; the same computation applies to the pair (2). The stationary law of (Θ, I) has densities p_0, p_1 satisfying $p_0' = r(p_1 - p_0)$ and $(v_1 p_1)' = r(p_0 - p_1)$, with constant positive flux $J = p_0 + v_1 p_1$ and $\int_0^\pi p_0 = \int_0^\pi p_1 = \frac{1}{2}$; then

$$\lambda(r) = -\frac{3}{4} + 2\int_0^\pi \cos(2\theta)\,p_1(\theta)\,d\theta.$$

With $H(\theta) = \theta - \frac{1}{4}\log|\tan\theta|$ (so that $H' = 1/v_0 + 1/v_1$) the solution is explicit on each quarter-circle: for $0 < \theta < \pi/2$,

$$F_-(\theta) = e^{-rH(\theta)}\int_\theta^{\pi/2} e^{rH(s)}\,ds, \qquad p_0 = J(1 + rF_-), \qquad p_1 = \frac{JrF_-}{2\sin 2\theta},$$

and for $\pi/2 < \theta < \pi$,

$$F_+(\theta) = e^{-rH(\theta)}\int_{\pi/2}^\theta e^{rH(s)}\,ds, \qquad p_0 = J(1 - rF_+), \qquad p_1 = -\frac{JrF_+}{2\sin 2\theta},$$

with J fixed by normalization. On the second branch $H' > 1$, which gives $rF_+ < 1$ and $p_0 > 0$. We evaluate these integrals as ordinary differential equations in the coordinate $x = \log|\tan\theta|$ on $(0, \pi/2)$ and $x = -\log|\tan\theta|$ on $(\pi/2, \pi)$, so that $\cos 2\theta = -\tanh x$ on the first branch and $\tanh x$ on the second, which resolves the power singularities at the attracting and repelling directions of A_1 (`code/stationary_density.py`). A second program (`code/stationary_density_second.py`) integrates the same equations with a different integrator, normalizes by $\int_0^\pi p_0 = \frac{1}{2}$ and integrates $(\cos 2\theta - 1)p_1$; its values of λ differ from these by less than 10^{-12} at $r = 1/10$, 5 and 20. Table 2 compares the computation from the stationary density with two other estimates at those rates.

On a grid of 403 rates in $[10^{-3}, 10^2]$ the computation from the stationary density gives exactly two sign changes, near $r = 1.2228$ and $r = 7.7616$ (to ten places 1.2228361663 and 7.7615867624; Figure 1). A grid of 151 rates in $[10^{-3}, 10^3]$ gives the same two sign changes. The two sign changes and the absence of others are numerical observations. As $r \to \infty$, $\lambda(r)$ tends to the largest eigenvalue $-\frac{3}{4} + \frac{\sqrt{3}}{2} \approx 0.1160$ of $\bar{A}$; this follows from [4, Proposition 6.4], whose condition (C) [4, Definition 6.3] holds because the lower eigenspace of $\bar{A}$ is a line and

r	stationary density	Markov chain, 16384 cells	Monte Carlo (standard error)
0.1	0.216444855966	0.216417929898	0.21600 (0.00036)
5	−0.040611505820	−0.040711536760	−0.04159 (0.00066)
20	0.077660185766	0.077491305516	0.07702 (0.00061)

TABLE 2. Three estimates of $\lambda(r)$. The Markov chain discretizes the direction into cells with upwind moves at rate $|v_i|/\Delta\theta$; its remaining bias is mesh error. The Monte Carlo runs use exact matrix flows between exponential holding times, two seeds of 2×10^6 cycles each; the standard errors are batch estimates.

A_0 leaves no real line invariant (their redraw intensity μ is $2r$ in our notation). With the limit $\frac{1}{4}$ as $r \to 0$ (Proposition 4.1), both ends of the curve are unstable and the exponent dips below zero in between.

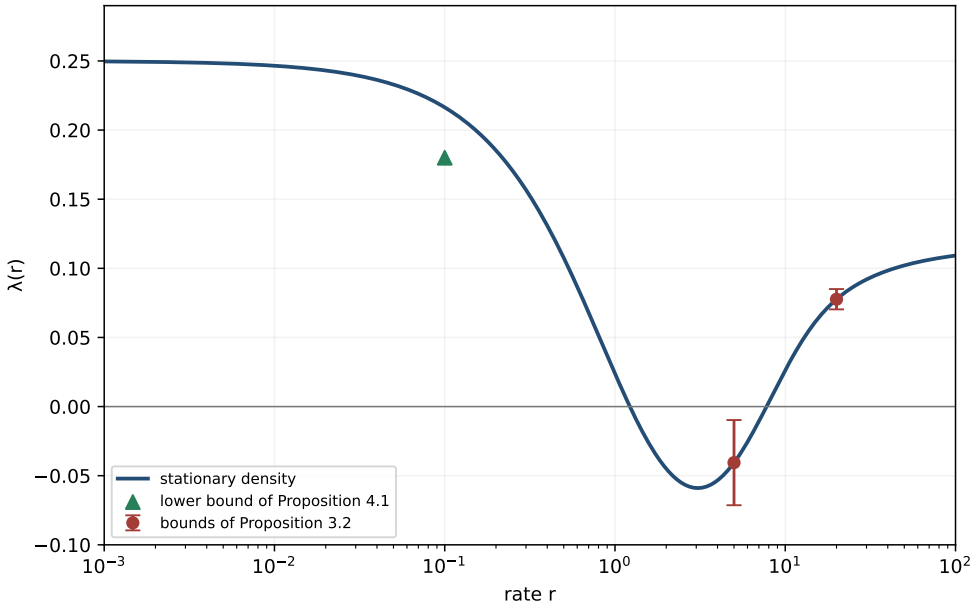


FIGURE 1. The exponent $\lambda(r)$ computed from the stationary density, on a logarithmic rate axis. The marked points are the rates where the sign is proved.

6. The Hurwitz case

In the example of [8] that precedes Open question 3 both matrices are Hurwitz. Whether two Hurwitz matrices can have a set of unstable rates with two components is open.

Proposition 6.1. *If A_0 and A_1 are normal and Hurwitz, there is $\epsilon > 0$ such that $|X_t| \le e^{-\epsilon t}|X_0|$ for every $r > 0$ and every path of I. In particular $\limsup_{t\to\infty} t^{-1} \log \|\Phi_t\| \le -\epsilon$, and no rate is unstable.*

Proof. A normal matrix has symmetric part with eigenvalues equal to the real parts of its own eigenvalues, so a normal Hurwitz matrix A satisfies $x^{\mathsf{T}} A x \le -\epsilon|x|^2$ for some $\epsilon > 0$. With ϵ the smaller of the two constants, $\frac{d}{dt}|X_t|^2 \le -2\epsilon|X_t|^2$ along every switching path, so $|X_t| \le e^{-\epsilon t}|X_0|$. □

Both matrices of (2) are normal, so no pair $A_0 - c_0\mathrm{Id}$, $A_1 - c_1\mathrm{Id}$ obtained from (2) is a Hurwitz pair with an unstable rate. A Hurwitz pair whose set of unstable rates has two components must therefore contain a non-normal matrix, as the pair of [8] does.

Among Hurwitz pairs obtained by shifting random traceless pairs, none has two intervals of unstable sampled rates. The program `code/search_ctmc.py` draws 600 pairs of traceless planar matrices, 200 of each of three kinds: A_0 the generator of the rotations, or $A_0 = \mathrm{diag}(1, -1)$, with A_1 the traceless part of a matrix with integer entries in $[-15, 15]$ times a random scale; and both matrices the traceless parts of matrices with integer entries in $[-10, 10]$. For each pair λ is computed at 49 rates in $[10^{-3}, 10^{3}]$ from a 256-cell discretization of the direction. For 85 pairs this computation fails at some rate (a singular linear system or a non-finite value), and these pairs are dropped; 515 remain, 194, 169 and 152 of the three kinds. Each matrix is shifted by a multiple of the identity to spectral abscissa $-\gamma$, $\gamma \in \{0.01, 0.05, 0.1\}$, which lowers the curve λ by a constant. Of the 1545 shifted pairs, 1350 have no unstable rate, 195 have one interval of unstable sampled rates, and none has two (`data/hurwitz_search.json`). Components narrower than the spacing of the rates are not resolved.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [16] and working together inside the Brain framework [18], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [17], the Math Brain repository [16] and the Brain framework repository [18].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [18] and Math Brain [16]. A Codex agent (OpenAI) found the pair (2), constructed the certificates and proved Theorem 1.1; a second Codex agent checked the proofs and recomputed the certificates, the stationary density and the Monte Carlo estimates with its own programs; a Claude agent (Anthropic) computed the stationary density with its own discretization and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

The certificates, the programs that check them and the numerical computations are in the repository

`https://github.com/yspennstate/planar-switching-two-instability-intervals`.

The exact checks need only the Python standard library; a recomputation of the bounds (8) in outward-rounded interval arithmetic (`code/interval_bounds.py`) uses `mpmath` [13], and the numerical computations use NumPy [6] and SciPy [19].

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Appendix A. Notation

Symbol	Meaning	First appears
X_t, I_t	the planar component and the switching state	Section 1
A_0, A_1	the two matrices; from Theorem 1.1 on, the pair (2)	Section 1
r	the rate of each jump of I	Section 1
Φ_t, $\lambda(r)$	the fundamental matrix and the top Lyapunov exponent	Section 1
U	the set of unstable rates	Theorem 1.1
$\bar{A}$	$(A_0 + A_1)/2$	Section 1
Θ_t	the direction of X_t modulo π	Section 1
h_i, v_i	radial growth rate and angular velocity in state i	Section 1
$f = (f_0, f_1)$, g_i	a pair of C^1 functions of period π and $h_i + v_i f_i' + r(f_{1-i} - f_i)$	Section 1
K	the number of harmonics in a certificate	Section 1
u_θ, $u_\theta^\perp$	the unit vector at angle θ and its rotation by $\pi/2$	Section 2
h_A, v_A	radial growth rate and angular velocity of a matrix A	Section 2
R_t	$\lvert X_t \rvert$	(3)
$\mathcal{L}_r$	generator of the angular process (Θ, I)	(4)
T_0, T_1	holding times in states 0 and 1 during one cycle	Proposition 2.1
κ	$\max_i \lVert A_i \rVert$	Proposition 2.1
$\Theta^{(n)}$	the direction at the start of the n-th cycle	Proposition 2.1
k_r	the density of T_0 modulo π	(6)
m_r, $\overline{m}_r$	lower and upper bounds of the density (6)	Proposition 2.1
C_i, $\alpha_{i,k}$, $\beta_{i,k}$	Fourier coefficients of g_i	(7)
$c_{i,k}$, $s_{i,k}$	coefficients of f_i	Section 3
$f^{(5)}$, $f^{(20)}$	the certificates at $r = 5$ and $r = 20$	Proposition 3.2
p_0, p_1, J	stationary densities of (Θ, I) and their flux	Section 5
H, $F_\pm$	the functions in the explicit stationary density	Section 5
x	the coordinate $\pm \log \lvert \tan\theta \rvert$ for the density computation	Section 5
γ	minus the spectral abscissa in the search	Section 6

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel